Program: **BI Analytics**

Sprint: ***5.- Storytelling with Data***

Project: **Superstore – Order Return Causes | ANALYSIS REPORT**

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Overview

This analysis aims to determine the actions to be taken after appropriate measures and graphic demonstrations have revealed the reasons why the company experiences a considerable level of post-sales order returns.

Research

Summary of Analysis

To determine the causes of order returns, we attempted to cross-reference customer feedback with the available data. Unfortunately, Superstore's data does not provide information on the reasons customers cite for order returns, whether due to quality issues or operational deficiencies in shipping and/or delivery. Therefore, we will attempt to determine the causes based on statistical data. The following is the scope of the research:

Period: Four years (2018-2021), with annual and monthly comparative periods.

Location: US (50 states), covering all states.

Strategy: Comparative analysis of transactions (order returns) by product categories and sub-categories. We have also identified the market segments involved and even conducted an analysis at the individual customer level.

Measuring Returns: To obtain a true understanding of the level of order returns, we consider the following measures appropriate:

- Order Return Rate
- Number of Returned Orders
- Cost of Returned Orders

Causes of Returned Orders: The increase in returns is driven by a combination of:

- Product Issues
  Interpretation:
  Inconsistent sizing standards

Quality variability from suppliers
Inaccurate or incomplete product descriptions

- <u>Logistics Issues</u>
  Interpretation:
  Packaging not suitable for long-distance shipping
  Carrier handling issues
  Incorrect item fulfillment
- <u>Customer Expectation Gaps</u>
  No Information available

Overview of Visualizations

To display the analyzed information and draw conclusions, we have created the following Worksheets, Dashboard and Story in Tableau:

- Return Rate
- Correlation: Sales vs Returns by Sub-Category
- Return Rate by Customers
- Return Rate by Category
- Order Returns by Year/Month
- Return Rate by State
- Return Rate by Year/State
- ***Understanding Returns (Dashboard)***
- ***Presentation - Story Arc (Story)***

The following is the explanation of what each worksheet contains and how it should be interpreted.

All our worksheets and the dashboard itself are sensible to the following filters: *Year*, *Month*, *State*, *Product Category* and *Product Sub-Category*.

- <u>Return Rate</u>

  The top right corner of our dashboard shows the return rate in *US (50 states)* during the whole *period (2018-2021)* including all products *Categories and Sub-categories*. This rate will vary if we apply one, some or all the mentioned filters. It is the result of the calculation of the average number of returned orders.

- <u>Correlation: Sales vs Returns by Sub-Category</u>

Expressed in US$ using the product sub-category level to get a better idea of what kind of products are being returned.

- Return Rate by Customer

  This visualization displays the number of orders returned by individual customers, offering the ability to apply the aforementioned filters in various combinations. This functionality allows us to identify behavioral trends based on a variety of different factors.

- Return Rate by Category
  This visualization displays the number of orders returned by product categories, offering the ability to apply the aforementioned filters in various combinations. This functionality allows us to identify the product categories returns based on a variety of different factors.

- Return Rate by State
  Provides a graphical, macro-level overview of return patterns across each state.

- Return Rate by Year/Month
  This chart enables the analysis of return trends based on seasonality, as well as comparative analysis across specific months and years.

- Return Rate by Year/State
  Provides an annual, macro-level overview of return patterns across each state.

- Understanding Returns **(Dashboard)**

  How to Use a Dashboard

  First and foremost, the Dashboard should be viewed as a diagnostic tool designed to assist stakeholders in performing the following key functions:

  1. Monitor Return Performance
     a. Across various dimensions, including *Time*, *Geography*, and *Product Type/Quality*.

2. Investigate the Underlying Causes of Returns
   a. By utilizing all available filters to generate in-depth metrics, users can isolate specific issues by cross-referencing data points simultaneously to draw meaningful comparisons.

3. Make Data-Driven Decisions Based on the Results.
   a. To make well-founded decisions, stakeholders should begin with a general, global overview of the return rate and ask themselves whether the current level of returns is minimally acceptable.
   b. Regardless of that initial assessment, the *Time Dimensions* help determine during which periods or seasons return rates tend to rise.
   c. In the meantime, the *Geographic Dimensions* reveal which regions of the country exhibit the highest volume of returns.
   d. *The Product Type/Quality Dimensions* indicate which products have a higher propensity to be returned, while combinations of all these dimensions provide a deeper and more comparative insight into the identified problems.

- Presentation – Story Arc **(Story)**

Story Points: The following are the story points you will find in our Tableau Story, they will be supported for worksheets

1 — The Most Important Signal

> **Caption:** Between 2018-2021 this was the level of Order Return Rate at Superstore in United States including all products categories and sub-categories.

2 — How We Measure Returns

> **Caption:** Return rate shows customer behavior; total returns and cost of returns quantify operational impact. Use all three for a complete view.

3 — Geographic Hotspots

> **Caption:** California, Texas, and Florida show the highest return rates, driven by long shipping distances and heavy online ordering.

4 — Product Categories Driving Returns

Caption: In general, the three main product categories show the same return rates, however, Technology is the one who has the highest rate of returns.

5 — Customer Behavior Patterns

Caption: Filtering out one-time buyers reveals repeat customers with consistently high return rates, indicating expectation gaps.

6 — Composite View: Connecting the Factors

Caption: Combining geography, product category, and time reveals a chain reaction: inaccurate info + weak packaging + long-distance shipping.

7 — How to Use the Dashboard

Caption: Start with overall return rate, drill down by category or geography, and use filters to isolate root causes.

8 — Recommended Actions

Caption: Improve product descriptions, standardize sizing, reinforce packaging, retrain fulfillment centers, and monitor returns in real time.

9 — Conclusion & Next Steps

Caption: Addressing product clarity, quality consistency, and operational gaps will reduce returns and improve customer experience nationwide.

Conclusion

The rise in returns is **actionable and reversible**. By addressing product clarity, quality consistency, logistics reliability, and customer expectations, the company can significantly reduce return rates and improve profitability across all 50 states.